

Auditing Frame-Level AUC in Weakly Supervised Video Anomaly Detection: Granularity, Resolution, and Scene Bias

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GitHub Code

Abstract. Frame-level area under the ROC curve (AUC) is the dominant evaluation metric for weakly supervised video anomaly detection (WSVAD). Its standard form measures whether an anomalous frame out-ranks a normal frame drawn from anywhere in the test set. We refer to this comparison as *pooled AUC*, since it aggregates frame pairs across test videos regardless of source. Pooled AUC therefore credits both event localization and differences between video sources. We audit this protocol on UCF-Crime [20] across recent state-of-the-art models spanning different backbone families. Holding each model’s frame scores fixed, we read them under three pairing granularities: global, per anomaly category, and within each video, then repeat the same three-granularity readout on zero-shot scores computed from the models’ internal representations. We assess ranking reliability with a paired video bootstrap. Three findings follow. First, pooled AUC does not reliably predict within-video anomaly localization: models with similar pooled scores exhibit large localization differences and rank reversals under stricter granularities. Second, at the benchmark’s test-split size, pooled AUC lacks the resolution to support state-of-the-art margins reported in the field. Within each backbone family, it resolves no comparison at those margins, while within-video AUC resolves several over identical predictions. Learned representations further reveal that within-video anomaly structure and detector localization are decoupled. Third, on normal footage alone, every model we examine separates videos by recording properties, such as resolution and color encoding, indicating that scene sensitivity is shared across the setting rather than specific to any architecture. We publicly release a granularity-aware protocol computable from existing predictions and scene-factor annotations for UCF-Crime.

Keywords: Video anomaly detection · Weak supervision · Evaluation metrics · Dataset bias · Representation auditing

1 Introduction

Weakly supervised video anomaly detection (WSVAD) aims to localize anomalous events in long videos while using only video-level labels during training [20,

21,23]. Progress on VAD benchmarks [20,25] is commonly summarized by a single metric: frame-level anomaly prediction, evaluated by the ROC-AUC (area under the true positive rate vs. false positive rate curve) pooled across all anomalous-normal frame pairs in the test set. AUC is threshold independent and admits a useful ranking interpretation statistically. However, its statistical consistency concerns *precision* at a given sample size, and does not guarantee that the *target objective* is being measured [8, 11, 19]. A consistent estimator may still be too imprecise to rank at a finite sample n , or can be faithful but estimating the wrong quantity.

A WSVAD model scores positively in an AUC comparison whenever an anomalous frame receives a larger anomaly score than a normal frame, regardless of whether the ordering follows the event semantics, behavior, or the footage source. This exposes two distinct failures, indistinguishable by the metric. First, a high AUC may not imply genuine anomaly detection. As the credited comparisons are overwhelmingly cross-video, a model’s predictions can be rewarded by AUC while learning to separate recordings rather than actual anomaly events. At deployment, a detector must rank anomalous intervals above normal ones from the same camera stream. Thus, a model relying on cross-video separation might produce constantly elevated scores, permanent false alarm, on anomaly-associated cameras while missing anomalies in atypical contexts. Pooled AUC cannot distinguish such failure from true localization. Second, the cross-video evaluation protocol yields an AUC measure too coarse to support model ranking reliably. As the easy cross-video pairs dominate the pool, they compress most models AUC toward a common narrow range, which may fall within the metric’s own sampling error [4, 7].

In this work, we investigate the evaluation process of WSVAD models. Using the same frame-level predictions produced by existing methods, we ask: (i) what does the standard pooled evaluation actually measure, and does it resolve models ranking reliably? (ii) Does the anomaly signal dominate in each model’s learned representation, and to what extent do the final anomaly scores depend on it? (iii) What determines anomaly score responses in entirely normal video segments? (iv) Does the pooled AUC ranking hold when comparisons are restricted to frames from the same recording? To answer these questions, we make the following contributions:

- A multi-granularity evaluation framework that reads the same predictions under different pairing rules, disentangling cross-recording based separation from within-video anomaly localization. We apply this to both final scores and training-free prototypes constructed from learned representations.
- A resolution analysis of WSVAD benchmarks: by resampling test videos, we test which pairwise model comparisons are statistically distinguishable at each granularity. We reveal that standard pooled AUC cannot separate models at the sub-point margins the field reports, whereas within-video AUC can.

- A scene-reliance audit of over thirteen annotated binary scene factors, computed on normal frames only. We show that every examined model separates frames by recording properties in the absence of anomalies.
- We publicly release (i) the scene-factor annotations for the full UCF-Crime dataset, and (ii) code for an evaluation protocol computable from existing predictions.

2 Related Work

Weakly supervised video anomaly detection. Since the introduction of the large-scale UCF-Crime benchmark [20], weakly supervised video anomaly detection has evolved rapidly. Models have progressed from basic temporal instance selection to increasingly sophisticated mechanisms, including robust normality modeling, memory-augmented representations, and multimodal vision-language alignments [1, 10, 14, 21–24]. Across all works, methods are still predominantly ranked by pooled AUC on UCF-Crime, which is the protocol under question in this study. Rather than proposing a new VAD model, we examine whether this evaluation protocol supports the temporal-localization conclusions drawn from the same models and predictions.

Evaluation protocols for video anomaly detection. Limitations of frame-level aggregation have previously been identified from several complementary perspectives. Ramachandra and Jones [15] introduced region-based and track-based detection criteria because frame-level AUC rewards a video frame whenever any score exceeds the threshold, without establishing that the anomalous object or trajectory was correctly localized. Their analysis exposes a spatial credit-assignment problem, where the rank statistic may be computed correctly while the set of credited detections does not correspond to the intended task. Liu *et al.* [9] identify three limitations of current VAD evaluation, which are the reliance on a single-perspective temporal annotations, failure to reward early anomaly detection, and the inability of existing benchmarks to reveal scene overfitting. They address these limitations with annotation-averaged AUC/AP, Latency-aware Average Precision (LaAP), and hard-normal benchmarks constructed to test whether models generalize beyond anomaly-associated scenes. Rashvand *et al.* [16] focus on the limitation of frame-level evaluation, where it treats a video as a collection of isolated frames, while an anomaly is a coherent temporal event with an onset and duration. The authors introduce an event-centric protocol based on temporal-IoU event matching and multi-threshold event-level F_1 .

AUC and VAD model comparison. AUC estimates the probability that a randomly selected anomalous segment receives a higher score than a randomly selected normal segment. Its threshold independence and relative stability are useful, however, pooled AUC can obscure the operating region and anomaly distribution of practical interest [19]. Defenses of AUC often conflate two distinct properties: (1) the *prevalence invariance*, which represents the positive-negative

ratio within the estimand [8, 11], and (2) the *sampling precision* of the estimator at finite test size n . Only the former is granted by AUC’s ranking interpretation. The latter was quantified analytically by Hanley and McNeil [7, 12], showing that the sampling standard errors scale as $\frac{1}{\sqrt{n}}$ in the effective sample size, regardless of positive-negative sample balance. The effective samples size in temporally correlated data, such as video frames, is closer to the number of videos than the number of frames. At the scale of current WSVAD test sets, the standard sampling error might be order of magnitudes larger than the margins by which SoTA is currently claimed. Our video-level bootstrap verifies this empirically.

Benchmark, scene, and representation auditing. A parallel line of work asks whether benchmark performance reflects the intended evidence rather than dataset-specific shortcuts [5, 6, 17, 26]. Mumcu *et al.* [13] question whether multi-scene anomaly recognition is appropriately formulated, arguing that scene and anomaly semantics can become entangled under current benchmarks. This concern motivates our scene-reliance audit, but our analysis is performed at the evaluation level.

3 Granularity-aware VAD Evaluation

Let $s_i \in \mathbb{R}$ be the anomaly score of frame i , $y_i \in \{0, 1\}$ its frame-level label, v_i its video identity, and c_i the anomaly category of its video. We compute AUC at three granularities by changing the anomaly score source, and the set of eligible positive-negative frame pairs.

3.1 Prediction-Score Granularities

We evaluate a model’s frame-level scores under three pairing rules for AUC that differ only in which anomalous–normal frame pairs are eligible for comparison: *global* (any two videos), *per-category* (same anomaly category), and *within-video* (same recording). For all, models scores are fixed and we compute the same ROC-AUC statistic varying only the population. We use *global* and *pooled* interchangeably for the standard protocol.

Global AUC. The standard frame-level protocol established in the WSVAD field [20], which pools the full test set frames:

$$\text{AUC}_{\text{global}} = \Pr(s^+ > s^-), \quad (1)$$

where $\Pr(\cdot)$ denotes probability under a uniform random draw of one anomalous frame score s^+ and one normal frame score s^- from the frame pool in the test set. Since the anomalous and normal frames may originate from any video; behavioral and contextual separability are both rewarded in $\text{AUC}_{\text{global}}$.

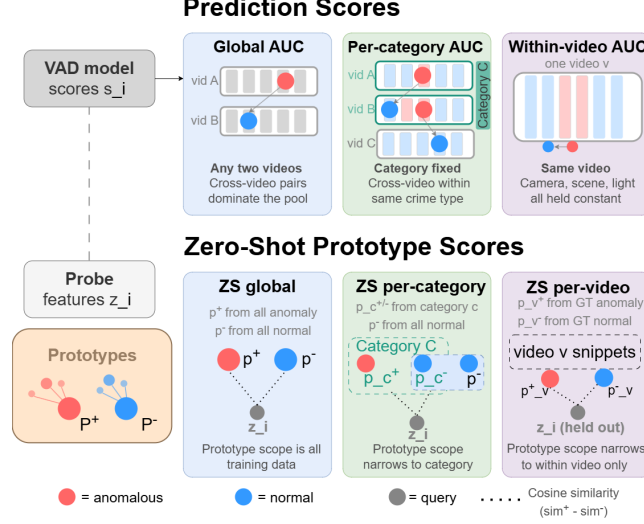


Fig. 1: Two score sources for AUC under three pairing rules. Each scoring mechanism ranks anomalous (red) against normal (blue) frames, differing only in which pairs are eligible. Top: the detector’s own prediction scores s_i . Bottom: prototype scores, where the pre-classification head embeddings from frozen VAD models are applied to construct a normal and abnormal class-mean (prototype). Then, an anomaly score for a snippet feature z_i is computed by its similarity to anomalous versus normal prototypes.

Per-category AUC. For anomaly category c , positives are the anomalous frames of videos labelled c , and negatives are the normal frames of those same videos; normal frames inherit the category of the video they occur in, and videos labelled Normal contribute nothing at this granularity. A single AUC is computed over each category’s full frame pool. The reported value averages these across categories,

$$\text{AUC}_{\text{cat}} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{AUC}(\{s_i : y_i = 1, c_i = c\}, \{s_i : y_i = 0, c_i = c\}), \quad (2)$$

so that each category contributes equally regardless of how many videos or frames it contains. Fixing the category constrains the type of event that the positives depict, but leaves the recording context free. Specifically, an anomalous frame from one video is still ranked against a normal frame from another. This granularity is therefore stricter than global pooling with respect to category composition, while still admitting cross-video comparisons.

Within-video AUC. Let $\mathcal{V}_{\pm}$ contain test videos with at least one anomalous and one normal frame. We compute an AUC inside every video and macro-average

across videos:

$$\text{AUC}_{\text{wv}} = \frac{1}{|\mathcal{V}_{\pm}|} \sum_{v \in \mathcal{V}_{\pm}} \text{AUC}(\{s_i : y_i = 1, v_i = v\}, \{s_i : y_i = 0, v_i = v\}). \quad (3)$$

The compared frames share the same recording, camera, location, and broad acquisition conditions. The model must rank the anomalous interval above the video’s own normal footage. We therefore interpret AUC_{wv} as a *temporal localization measure*, not as a replacement for global detection performance.

3.2 Zero-Shot Representation Readout

Prediction scores characterize the final detector, but cannot identify whether a failure originates in the representation or the scoring head. We therefore repeat the same three-granularity readout using zero-shot prototype scores on the feature immediately before the model’s final classification head.

Let $z_i \in \mathbb{R}^d$ represent the feature vector for snippet i . For a given granularity level g , denote the anomalous and normal support sets by $\mathcal{S}_g^+$ and $\mathcal{S}_g^-$, respectively. Their corresponding prototypes are defined as the mean feature vectors of each class

$$p_g^+ = \frac{1}{|\mathcal{S}_g^+|} \sum_{j \in \mathcal{S}_g^+} z_j, \quad p_g^- = \frac{1}{|\mathcal{S}_g^-|} \sum_{j \in \mathcal{S}_g^-} z_j, \quad (4)$$

where $p_g^+, p_g^- \in \mathbb{R}^d$. We score query feature z_i by its difference in cosine similarity to the two prototypes:

$$q_i^{(g)} = \text{sim}_{\cos}(z_i, p_g^+) - \text{sim}_{\cos}(z_i, p_g^-), \quad \text{sim}_{\cos}(a, b) = \frac{a^\top b}{\|a\|_2 \|b\|_2}. \quad (5)$$

The support sets are constructed for each granularity g : globally, per anomaly category, or within the query video. Per-category and within-video prototypes are constructed with knowledge of ground truth snippet labels, whereas global prototype is based on video labels only. For within-video prototype, the evaluated query snippet is excluded from its support set through a leave-one-out procedure. We evaluate the resulting scores $q_i^{(g)}$ using Equations (1)–(3).

3.3 Resolving WSVAD Model Comparisons

The multi-granularity analysis asks whether pooled AUC reflects the intended within-video localization ability, both for the detector’s final scores and for prototype readouts of its learned representation. A separate question is whether the margins observed at any granularity are large enough to support the resulting model ranking. For such question, it is worth distinguishing two properties. First, AUC is a statistically well-understood metric that is relatively stable under

repeated sampling and invariant to class prevalence in its pairwise interpretation [8]. This stability concerns how *precisely* AUC estimates a *fixed* comparison population. Second, statistical stability of an estimator does not imply that a finite benchmark has the resolution to reliably distinguish models with scores differing by only fractions of a point. In most VAD benchmarks where the test set is finite in terms of unique video scenes, the variation of AUC estimates under different video resamples can be several magnitudes wider than the margins between state of the art estimates.

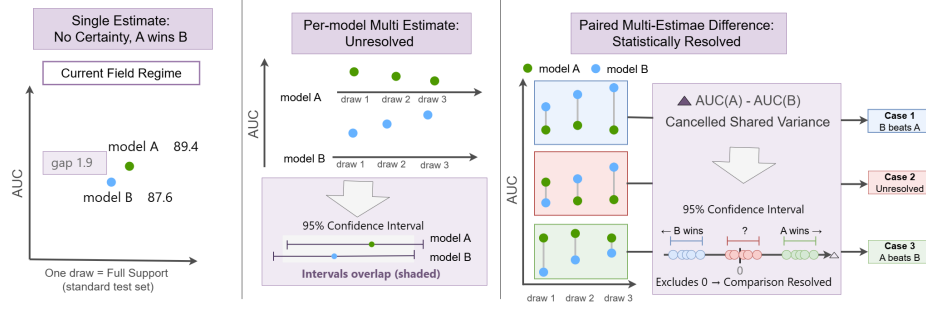


Fig. 2: AUC ranking resolution. Left: a single AUC per model. Middle: separate bootstrap sample intervals overlap, leaving comparison unresolved. Right: the difference resolves the ranking as it removes the shared effect on the video resamples.

We quantify the AUC uncertainty by repeatedly drawing evaluation videos with replacement, retaining all frames of each selected video, and recomputing AUC on each of the N resampled sets. The 2.5th and 97.5th percentiles of the resulting AUC distribution form a 95% confidence interval. Such an interval quantifies uncertainty in one model’s AUC estimate, but when intervals across models are overlapping, it does not differentiate a model from another. Therefore, model ranking requires the confidence interval of the paired AUC difference, computed from identical video resamples for both models. Figure 2 contrasts these readings on an illustrative model pair. Our paired video bootstrap and AUC difference comparison is the nonparametric counterpart to the DeLong test [4], which compares two correlated AUCs analytically via U-statistic theory, under the samples independence assumption. Since frames within a video are strongly correlated in our context, we resample at the video level to preserve this dependence structure.

Mathematically, let $\mathcal{V}$ denote the set of evaluation videos with $n = |\mathcal{V}|$. For each resample $r = 1, \dots, B$, where B is the total number of draws, we draw n videos from $\mathcal{V}$ with replacement forming the video multi-set I_r . All frames of each selected video are retained, and every model is evaluated on the same I_r . Let $\text{AUC}_g^m(I_r)$ denote the AUC of model m at granularity g computed on

replicate I_r . For a model pair (A, B) , the paired AUC difference $D_{AB}^{(r)}$ is

$$D_{AB,g}^{(r)} = \text{AUC}_g^A(I_r) - \text{AUC}_g^B(I_r). \quad (6)$$

Its percentile 95% confidence interval is obtained from the empirical quantiles ($Q_{0.025}$, $Q_{0.975}$) of these resampled differences as

$$\text{CI}_{95\%}(D_{AB,g}) = \left[Q_{0.025}(\{D_{AB,g}^{(r)}\}_{r=1}^B), Q_{0.975}(\{D_{AB,g}^{(r)}\}_{r=1}^B) \right]. \quad (7)$$

A model pair comparison is resolved at granularity g when $0 \notin \text{CI}_{95\%}(D_{AB,g})$. For model set $\mathcal{M}$, we summarize granularity g by the number of resolved unordered model pairs:

$$C_g = \sum_{\{A,B\} \subset \mathcal{M}} \mathbf{1}[0 \notin \text{CI}_{95\%}(D_{AB,g})], \quad 0 \leq C_g \leq \binom{|\mathcal{M}|}{2}. \quad (8)$$

Thus, C_g measures how many model comparisons AUC can statistically settle at granularity g .

4 WSVAD Scene Reliance Audit

The three-granularity breakdown and the AUC resolution show whether a model benefits from cross-video comparisons and whether its ranking remains consistent. However, for any resolved model comparison, none of these evaluations quantify whether the performance gain is genuine (from the use of anomaly cues), or confound (from exploiting the recording properties that correlate with the anomaly events). We tackle this by assessing model bias when it scores predefined scene factors in normal video footage.

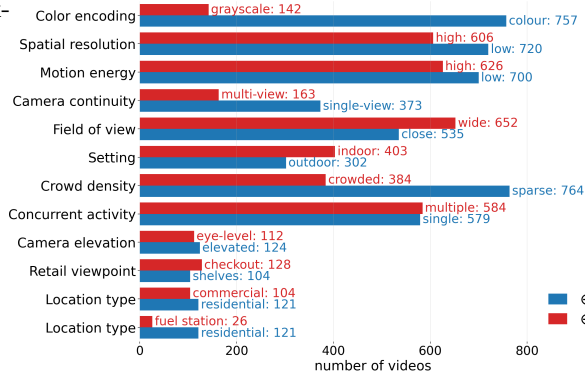


Fig. 3: Support of the scene-factor audit.

Scene factor annotation. We annotate the full UCF-Crime videos [20] with twelve binary scene factors covering acquisition conditions, viewpoint, environment, location, motion, and content composition. For each factor, a video is assigned to one of two poles ($\oplus$ or $\ominus$), or marked *ambiguous*. Ambiguous videos are excluded only from that factor, thus, each factor has its own evaluation support (Figure 3). Full annotation definitions and protocol details are provided in Supp. S1.

Table 1: Models assessed in this study. Feature backbones correspond to the original architectures adopted by the authors. AUC values are taken from the reported results, except for Universal MIL, where † indicates our implementation. Probe location is the representation immediately preceding the final linear classifier in each model.

Model	Venue	Probe location	Probe. dim	Feature backbone	AUC (%)
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	I3D	79.68
UR-DMU [23]	AAAI’23	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	86.97
BN-WVAD [24]	T-CSVT’24	Pre-classifier post-attention ($\tilde{\mathbf{x}}_{\text{att}}$)	512	I3D	87.24
PELAVAD [14]	TIP’24	Pre-classifier visual projection ($\mathbf{x}_v$)	704	I3D	86.76
π -VAD [10]	CVPR’25	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	90.33
GS-MoE [1]	ICCV’25	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	I3D	91.58
TEVAD [3]	CVPR’23	Pre-classifier fusion ($\mathbf{x}_v \oplus \mathbf{x}_t$)	2816	I3D + SwinBERT	84.92
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	VideoMAE v2 (ViT-g/14)	85.48
SST-WSVADL [2]	ECCV’26	Pre-classifier encoder ($\mathbf{x}_{\text{pre-cls}}$)	1024	VideoMAE v2 (ViT-g/14)	88.52
Universal MIL† [18]	CVPR’18	Pre-classifier hidden layer (pre- fc_2)	512	CLIP (ViT-L/14)	84.47
VadCLIP [21]	AAAI’24	Pre-classifier ($\mathbf{v} + \text{MLP}_2(\mathbf{v})$)	512	CLIP (ViT-B/16)	88.02
DSANet [22]	AAAI’26	Pre-classifier ($\mathbf{v} + \text{MLP}_2(\mathbf{v})$)	512	CLIP (ViT-B/16)	89.40
TrCLIP-VAD [18]	Neural Netw.’26	Pre-classifier ($\mathbf{f} + \text{MLP}_2(\mathbf{f})$)	1280	CLIP (ViT-L/14) + text	88.59

Measuring model bias. We compare model performance on normal frames only from different scene factor groups. Since each factor has two opposite poles, we quantify a model bias by its AUC ranking separation between two scene poles. We call this *bias-AUC*. As the anomaly label is constant across the comparison, any departure from 0.5 reflects a response to the scene factor rather than discrimination between anomalous and normal content. Writing $\bar{s}_v^0$ for the mean score over the normal frames of video v , the video-level *bias-AUC* for factor j is

$$b_j = \Pr(\bar{s}_{v_{\oplus}}^0 > \bar{s}_{v_{\ominus}}^0) + \frac{1}{2} \Pr(\bar{s}_{v_{\oplus}}^0 = \bar{s}_{v_{\ominus}}^0), \quad v_{\oplus} \sim \mathcal{V}_j^{\oplus}, \quad v_{\ominus} \sim \mathcal{V}_j^{\ominus}. \quad (9)$$

where $\mathcal{V}_j^{\oplus}$ and $\mathcal{V}_j^{\ominus}$ denote the sets of videos assigned to the two opposing poles of scene factor j . The *bias-AUC* is obtained as the ROC-AUC with pole $\oplus$ treated as the positive class. Values above 0.5 indicate higher normal-frame scores at positive pole, and below 0.5 indicates higher scores at the negative pole. We compute $d_j^{\text{bias}} = |b_j - 0.5|$ when comparing strength irrespective of sign.

Uncertainty and practical significance. We resample videos with replacement, stratified within each factor pole so that both are still represented. We keep each sampled video’s frames together to preserve temporal dependence, and recompute Equation (9) over B replicates. We report the percentile 95% interval and

$$p_{\text{bias}}^{(j)} = \frac{1}{B} \sum_{r=1}^B \mathbf{1} \left[\left| b_j^{(r)} - 0.5 \right| > 0.05 \right], \quad (10)$$

which is the fraction of replicates falling outside a practically-unbiased band of ± 0.05 . This distinguishes a factor with a small but consistent effect from one whose observed effect varies depending on which videos happened to be selected. We further comment on the effect of unequal pole sizes in Supp. S1.3.

5 Experimental Results

5.1 Dataset and Implementation Details

The headline analysis in this work emphasizes comparisons across recent top-tier WSVAD methods with different backbone families. Table 1 presents the models examined in this work. We adopt the official UCF-Crime [20] dataset for the three-granularity evaluation and scene-reliance analysis. We use the frame-level temporal annotations provided by [2]. For feature extraction, we adopt the same public I3D, VideoMAEv2, and CLIP features produced in previous works [2, 3, 10, 21, 23, 24]. Snippet size is set to 16 in all experiments. For CLIP-based variants, we substitute the center frame feature for the snippet. Global, per-category, and within-video scores are computed from the public pre-trained checkpoint predictions, except for few models remarked in Table 1, which we re-train. Prototype-based AUC refers to the AUC computed on probe features extracted from the frozen VAD models. Score-based AUC is calculated from the prediction scores of the VAD models based on the official ground truths, whereas Prob-AUC is computed on soft multi-annotator labels [9]. All AUC values are reported on the official UCF-Crime test set (290 video). The scene-factor audit in Section 4 has a larger support, including all videos from the train and test sets, to provide adequate scene diversity. The audit is descriptive rather than a performance estimate. No model is trained, selected, or calibrated on these annotations.

5.2 Anomaly Signal in Learned Representations

Table 2 (left-most column group) reports prototype-based AUC at the three granularities, computed without the model’s classification head. At the video granularity, the anomaly direction is built per video, leaving the query snippet out when constructing the normal and abnormal prototypes. Since ground truth frame labels are used to construct the prototypes, we treat AUC as a diagnostic ceiling on accessible signal, not as achievable detection performance.

The frozen backbones already contain a dominant within-video anomaly representation. VideoMAEv2 and I3D achieve a within-video ZS-AUC of 90.30 – 90.47, while CLIP reaches 88.34. Training WVAD models on these features does not consistently strengthen this signal. Compared to VideoMAEv2, within-video ZS-AUC decreases to 71.97 for UR-DMU, 74.11 for BN-WVAD, and 73.62 for SST-WSVADL. This indicates that the learned representations substantially attenuate the linearly accessible anomaly structure already present in the frozen backbone. In contrast, CLIP-based models exhibit a different behavior, improving the oracle ZS-AUC over the frozen CLIP backbone. These models retain or even strengthen a locally accessible anomaly direction, despite their lower global ZS-AUC. This suggests that the observed behavior is not a universal consequence of WSVAD training and therefore cannot yet be attributed to the choice of backbone, temporal receptive field, or architectural design.

Table 2: Prototype-based AUC vs. Score-based AUC vs. Prob-AUC at three granularities. Left: zero-shot prototype AUC on internal features. Middle: the same three granularities from each model’s final scores s . Right: the same three granularities under soft multi-annotator labels [9]. Backbone-only rows (VideoMAEv2, I3D, CLIP) have no Score-based AUC or Prob-AUC entry. Model ranks are marked as **Top-1st**, **Top-2nd**, and **Top-3rd**. [†]Indicates our re-training.

Feature / model	Prototype-based AUC			Score-based AUC			Prob-AUC		
	Global	Per-cat.	Within-video	Global	Per-cat.	Within-video	Global	Per-cat.	Within-video
VideoMAEv2	83.87	49.40	90.30	—	—	—	—	—	—
UR-DMU [†] [23]	89.58	54.59	71.97	88.08	76.50	75.83	89.51	76.80	79.40
PEL4VAD [†] [14]	49.24	49.82	48.00	87.56	73.78	76.80	89.26	73.12	76.77
BN-WVAD [†] [24]	84.75	51.29	74.11	86.76	70.74	70.44	88.76	71.13	74.10
SST-WSVADL [2]	74.72	54.56	73.62	88.52	78.11	75.75	89.48	77.88	78.67
GS-MoE [†] [1]	71.66	51.19	69.59	87.95	75.99	76.62	89.37	77.72	79.36
Pi-VAD [†] [10]	74.95	47.98	74.95	87.99	77.58	77.83	89.57	75.68	76.99
I3D	69.53	51.12	90.47	—	—	—	—	—	—
UR-DMU [23]	73.35	54.57	68.17	86.97	74.22	74.13	86.03	74.67	73.89
PEL4VAD [14]	68.27	56.42	86.94	86.76	73.22	75.42	86.87	71.54	73.07
BN-WVAD [†] [24]	73.62	55.51	71.44	84.01	65.37	72.10	85.65	66.25	70.79
SST-WSVADL [2]	75.36	60.54	68.98	86.50	74.46	73.73	85.76	74.85	73.33
GS-MoE [1]	76.02	56.35	70.65	91.58	83.86	86.79	89.26	80.25	82.59
Pi-VAD [10]	73.24	49.31	70.81	90.33	76.88	79.66	87.31	72.68	77.70
TEVAD [3]	77.47	48.02	88.79	84.92	66.53	70.79	84.61	66.25	70.79
CLIP	68.55	—	88.34	—	—	—	—	—	—
VadCLIP [21]	75.74	54.47	92.71	87.79	70.30	67.64	89.47	71.39	74.75
DSANet [22]	78.89	51.13	90.75	89.44	75.74	69.81	90.10	75.92	75.61
TrCLIP-VAD [18]	78.62	50.78	93.10	88.59	72.01	73.37	89.66	70.18	75.36

Across backbone families, per-category ZS-AUC remains close to chance, ranging from 47.98 to 60.54. This suggests that the anomaly might not be encoded as a distinct concept in the learned representations, resulting in poor transferability even across videos that belong to the same anomaly category. This holds even for models trained with explicit category-level supervision (VadCLIP and DSANet).

5.3 Representation Signal vs. Detector Performance

Section 5.2 measures the anomaly signal dominance within each model’s representations, whereas this section reads the models own anomaly scores at the same three granularities, reporting what their trained heads deliver (Table 2).

Representation signal does not predict localization. We observe that within-video score-based and prototype-based AUC disagree, and the disagreement reverses the model ranking (Table 2, *Within-video* prototype and score columns). The five models with the highest prototype AUC (86.94 – 93.10) deliver the *lowest* within-video score AUC (67.64 – 75.42). Conversely, the models with lower prototype AUC (48.00 – 74.95) deliver higher score AUC (75.75 – 86.79). The extremes make this observation concrete; PEL4VAD (VideoMAEv2) localizes at 76.80 from a prototype AUC of 48.00 (at chance), whereas VadCLIP reaches a 92.71 prototype ceiling but only 67.64 within-video score AUC. Thus, the readability of a video-specific anomaly direction from the representation appears

to be decoupled from the effectiveness of the scoring head in ranking anomalous against normal snippets in that video. This dissociation is not an artifact of the original temporal annotations. Under soft multi-annotator labels [9] (Table 2, Prob-AUC), absolute values and several ranks shift, yet the mismatch with Prototype-AUC persists. The effect is therefore robust to annotation variation on the score side.

Pooled AUC cannot separate the methods it ranks. Across all granularities in Figure S1, the per-model intervals overlap substantially. At the pooled granularity, every model’s interval contains every other model’s AUC estimate. As established in Section 3.3, overlapping intervals do not settle the models ranking. We therefore test the paired difference directly, computing Equations (6) and (8) on all models and count the number of resolved comparisons by the metric. Table 3 presents the resolution counts, and Table S4 presents detailed information on the resolved pairs.

Analyzing Tables 3 and S4 jointly, the overall resolved counts seem to overstate the AUC’s discriminative power. Among 34 resolved pairs across the model zoo, 25 are the easy cross-backbone comparisons, e.g. an I3D-based model against a CLIP-based one. Such comparisons are rarely informative, since a new method claims the state of the art by outperforming prior methods on the *same* features. Restricted to the same-backbone comparisons, global AUC resolves 0 of 15 VideoMAEv2-based pairs and 0 of 3 CLIP-based pairs, while the within-video readout resolves 5 and 1 of the same pairs. Even though the I3D family appears to be the exception (9 of 21 resolved), its resolved pairs are consistently centred on two models. The first is GS-MoE (I3D), which sits 3.5 points above the rest of the family, and the second is BN-WVAD (I3D), which sits 2.4 points further below. Among the five competitive I3D models, global AUC resolves zero pairs. Across all three families on UCF-Crime, pooled AUC seems to detect outliers separated by several points and resolves nothing at the margins the field publishes.

The stricter granularities also change which model wins. DSANet holds the highest pooled AUC of any CLIP-based detector (89.44), yet within video it is outperformed, with resolved intervals, by PEL4VAD (+7.04), GS-MoE (+6.93), UR-DMU (+6.06), SST-WSVADL (+6.04), and Pi-VAD (+7.99). None of latter models are ranked above DSANet by the pooled AUC (Table 2). The DSANet-PEL4VAD pair is unresolved at the pooled granularity and resolved within video, but in the *opposite* direction of the reported SoTA. Per-category, by contrast, largely preserves the pooled ordering among the stronger models, adding more resolutions that confirm the above pattern.

This bears directly on how improvements are claimed. DSANet is proposed as a refinement of VadCLIP and reports gains across every setting [22]. While

Table 3: Model pairs resolved under each granularity. Spread is max-min of point estimates in the subset.

Subset	Granularity	Pairs resolved	Spread (pts)
All (16 models)	Global	34 / 120	7.64
	Per-category	55 / 120	19.56
	Within-video	53 / 120	19.15
VideoMAEv2 (6)	Global	0 / 15	1.75
	Per-category	5 / 15	7.37
	Within-video	5 / 15	7.38
I3D (7)	Global	9 / 21	7.64
	Per-category	14 / 21	19.56
	Within-video	13 / 21	16.00
CLIP (3)	Global	0 / 3	1.65
	Per-category	1 / 3	5.44
	Within-video	1 / 3	5.73



Fig. 4: Scene reliance audit on normal frames only.

the two are separated per-category (+4.80, resolved), the global and within video differences are unresolved, and both models are outperformed by the same set of simpler detectors (Table S4). From observations in Table S4, no pair separated by a margin 0.4 – 0.5 AUC resolves in any backbone family, yet recent methods on UCF-Crime claim improvements within such margin [2, 14, 21, 22].

5.4 Scene Reliance of Anomaly Scores

When anomaly direction is not captured by the class-mean probe of learned representations, as shown in previous sections, the final scores may rely on anomaly-relevant evidence or on recording properties that co-vary with it. We directly

evaluate the scene bias, quantified as the deviation of the *bias-AUC* from the null band $p_{\text{bias}} \pm 0.05$ across the twelve labeled scene factors (Figure 4).

Bias to Recording Properties. Two acquisition factors are displaced in the same direction for every score sources. Colour encoding falls below 0.5 throughout (0.17 to 0.47) and spatial resolution above it (0.56 to 0.71); resolution reaches $p_{\text{bias}} = 1.000$ for every model. This unanimity spans all backbone families, architectures, and training objectives. Camera continuity, motion energy, and camera elevation follow the same pattern but at smaller magnitude. Field of view is the one acquisition factor with no bias response, where every model sits inside the practically-unbiased band (0.45–0.54).

Bias to semantic properties. Scene response is not confined to acquisition properties. Location type, crowd density, and concurrent activity displace most models above 0.55, while outdoor-indoor setting sits consistently below 0.5. These semantic and content biases are real but smaller in magnitude than the acquisition biases, and on most factors the models agree in direction. Only the location-type factors show disagreement in direction across models and backbones.

Backbone and head contributions. The bias response does not seem to originate in any trained detector. Universal MIL, a minimal head trained separately on VideoMAEv2, I3D, and CLIP features, shows similar directions on the same factors as the full models (Figure 4b). The bias magnitude shifts with the feature space more than with the model, therefore models that share a backbone cluster together. For example, the I3D columns show strong color-encoding response for GS-MoE (0.17) as well as for the minimal head (0.22). Thus, the backbone seems to set the regime, while model architectures and objectives shift values few points around it.

Impact of scene bias on anomaly detection. A scene-conditioned decomposition of AUC on anomaly-vs-normal frames shows that the measured bias rarely inflates the anomaly frames scores, except for the color encoding factor. The bias response is present in both anomaly and normal frame scores, thus, it mostly cancels in with frames pooling. See discussion in Supp. S1.2 and S2.2.

6 Conclusion

We asked what pooled frame-level AUC measures on WSVAD benchmarks, such as UCF-Crime, by reading the same anomaly scores under progressively stricter pairing rules and auditing what they respond to. Three key findings follow. First, our experiments on UCF-Crime show that the pooled AUC evaluation protocol lacks the discriminative power to support the rankings built on it. Second, the learned representation by WSVAD models and their predictions are decoupled. The models with the most dominant within-video anomaly structure may localize worst, whereas the strongest within-video detectors may produce representations that no class-mean probe can read. Pooled AUC registers neither of these

patterns. Third, every considered model separates normal footage by recording and semantic properties, such as resolution, colour encoding, and location, which carry no anomaly signal by construction.

We do not indict AUC as a statistic, since what fails mostly is the population of comparisons constructed for the pooled protocol. Dominated by frame pairs from different recordings, pooled AUC can reward separability without learning any anomaly concept. The correction is inexpensive and can be derived from the available predictions produced in previous studies. Model comparisons can be resolved by: (1) reporting within-video and per-category AUC alongside the pooled number, and (2) testing claimed improvements as paired differences on shared video resamples. We release our annotations and protocol to make this the path of least resistance.

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